

Previous: Harness-Bound Training

Single Harness



Claude Code



Bash



Read



Edit



Plan

Switch Harness



Openhands-SDK



Terminal



file_editor



task_tracker



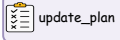
Codex



exec
commad



exec
command



update_plan



Ours: Hint—Harness Invariant Training

Real Trajectory



Code task



Terminal



Read



edit



Search



Plan

Harness Protocol



run_shell



read_file



edit_file



Search



track

same semantic
trajectory

Counterfactual Harnesses



Vocabulary

bash

excute_bash

terminal



Granularity

split

unified editor

shell-only



Schema & Docs



Observatisons

plain

line-numbered

truncated



Completion

finish tool

final reply

Harness Invariant
Agent



claude
code



Codex



OpenCode



Openhands